

Descriptive Analytics on Flipkart Sales Data

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Flipkart Sales Data Descriptive Analytics Report

Best-Sellers, Revenue Trends, Regional Demand, Payment Mix & Order Fulfilment Analysis

A full descriptive-analytics pass over 100 simulated Flipkart transactions spanning 7 categories, 44 products across 15 cities — covering revenue, product performance, geography, payment behaviour, order fulfilment and customer ratings, along with the architecture that would run this at real marketplace scale.

Metric	Value
Total Revenue	Rs. 602,883
Total Orders	100
Total Units Sold	160
Average Order Value	Rs. 6,029
Average Rating (Delivered)	4.35 / 5
Categories / Products / Cities	7 / 44 / 15

Report Date: August 2026 | Dataset: Flipkart Sales Dataset (Sample, PDF export)

1. Objective & Business Questions

This report builds a descriptive-analytics layer over Flipkart order data to answer six recurring business questions that any marketplace operations, category, or customer-experience team needs answered on a recurring basis:

- Which product categories and individual products generate the most revenue and order volume?
- Where is customer demand concentrated geographically, and how even is that spread across cities?
- How do customers pay — UPI, cards, EMI, Net Banking, or Cash on Delivery — and does that vary by category?
- How healthy is order fulfilment — what share of orders are Delivered versus Cancelled, Returned, or still Shipped?
- How satisfied are customers, based on ratings left on delivered orders?
- Which products carry disproportionate revenue weight (Pareto risk), and how does demand trend across the year?

Scope of this report

This is a descriptive analysis: it summarizes what already happened across the 100 orders in the dataset. It does not forecast future demand or recommend specific reorder quantities — see Section 12 for how this would extend into predictive territory.

2. Data & Methodology

2.1 Dataset Overview

The source file is a Flipkart order-level table containing one row per transaction: order ID, order date, category, product name, quantity, unit price, order total, delivery city, payment mode, order status, and customer rating. After parsing, the usable dataset comprises:

Metric	Value
Total transactions parsed	100
Categories	7
Unique products	44
Cities	15
Payment modes	6 (UPI, Credit Card, Debit Card, Net Banking, EMI, Cash on Delivery)
Date range	Jan 2025 – Dec 2025
Parse success rate	100% of rows (0 dropped)

2.2 Data Quality Notes

Unlike a typical spreadsheet export, this source file preserved clean, unambiguous columns for date, category, product, quantity, price, city, payment mode, and status — so no reconstruction or pattern-matching was required. The one field needing special handling was **rating**: this is only populated for orders in the **Delivered** state and appears as a dash ("-") for Cancelled, Returned, or Shipped orders, since a customer can only rate an order they actually received. Ratings analysis in Section 9 is therefore scoped to the delivered subset only, and this is called out wherever it applies.

2.3 Methodology Reference

Revenue = Unit Price x Quantity, summed per row (matches the Total column in the source file). Category- and product-level rollups use exact string grouping on the category and product fields. City- and payment-mode splits use exact grouping on their respective columns. Average Order Value (AOV) is computed as total revenue divided by order count within each grouping. All monetary figures are in Indian Rupees (Rs.).

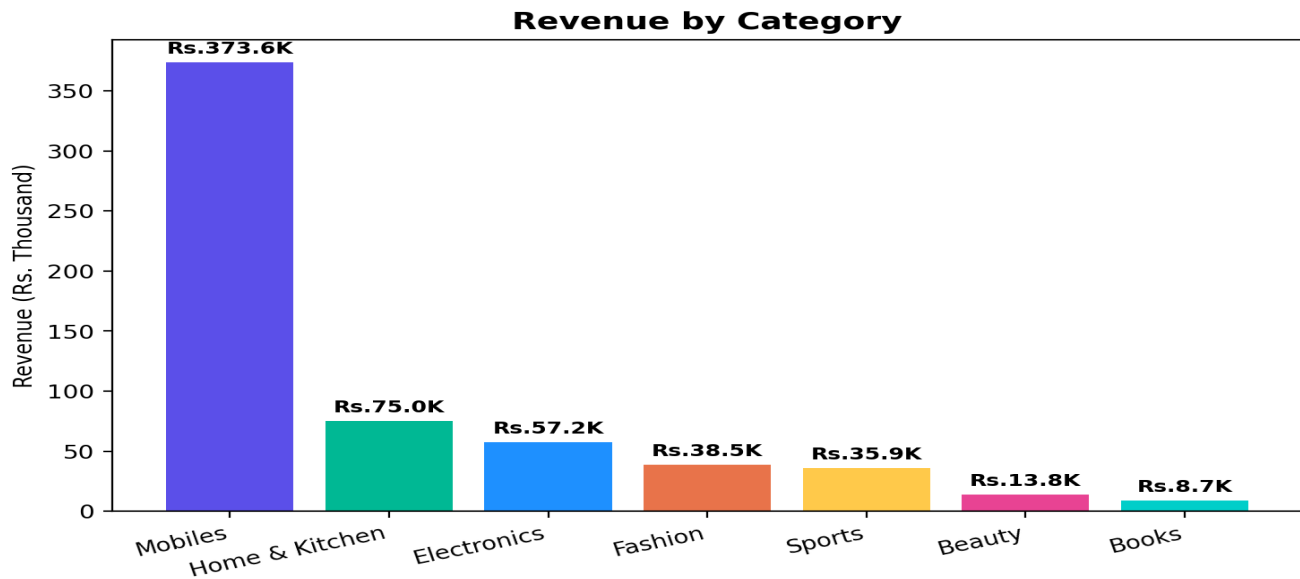
3. Summary of Key Metrics

Headline numbers across all 100 transactions in the dataset, before the section-by-section breakdowns that follow.

Metric	Value
Total revenue	Rs. 602,883
Total orders	100
Total units sold	160
Average order value (AOV)	Rs. 6,029
Delivered order rate	64% (64 of 100)
Average customer rating (delivered)	4.35 / 5
Top single product by revenue	Samsung Galaxy M34 (Rs. 105,328)
Top single product by units	Ikigai (10 units)
Products needed for 80% of revenue	15 of 44
Leading category	Mobiles (62.0% of revenue)
Leading city	Hyderabad (Rs. 100,030)
Dominant payment mode	Cash on Delivery (19% of orders)
Peak sales month	Jan (Rs. 136,286)

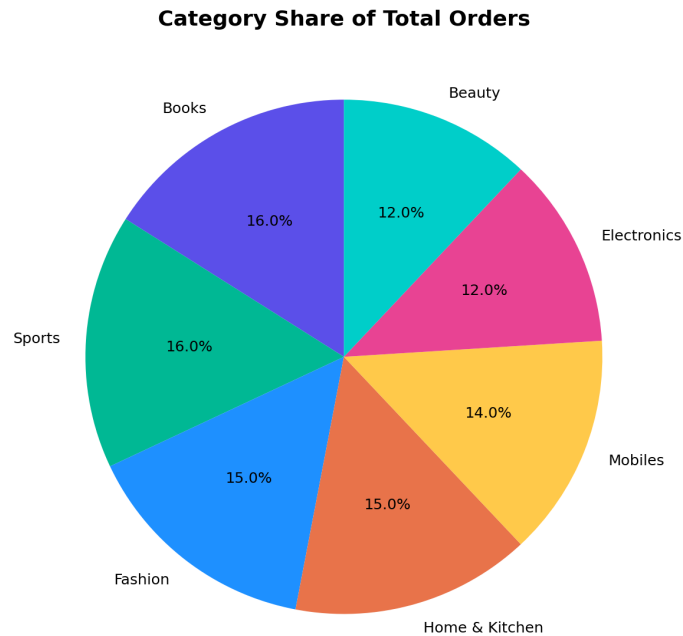
4. Revenue Analysis

4.1 Revenue by Category



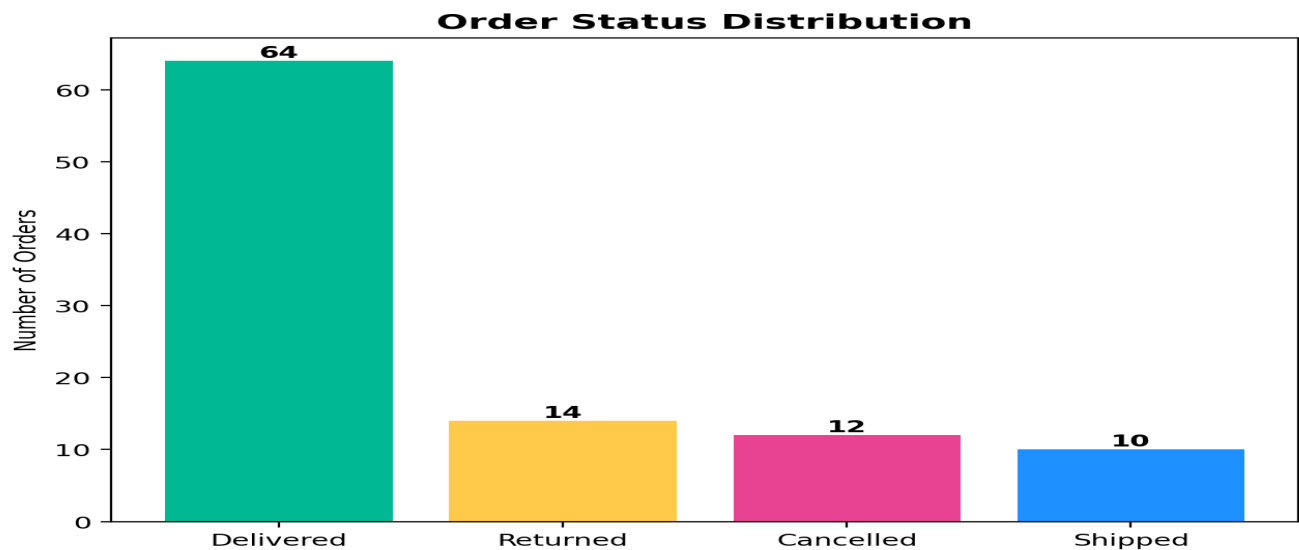
- **Mobiles** is the clear leading category with Rs. 373,608 in revenue — 62.0% of the total.
- **Books** trails the field at Rs. 8,722, an opportunity area for catalogue expansion or targeted promotion.
- The top two categories (Mobiles and Home & Kitchen) together account for 74% of total revenue, showing concentration in a handful of core categories.

4.2 Category Share of Total Orders



- **Books** drives the highest order volume (16 orders, 16% of all orders).
- Order share broadly tracks revenue share (Section 4.1), though categories with higher-ticket items like Mobiles contribute more to revenue than their order count alone would suggest.

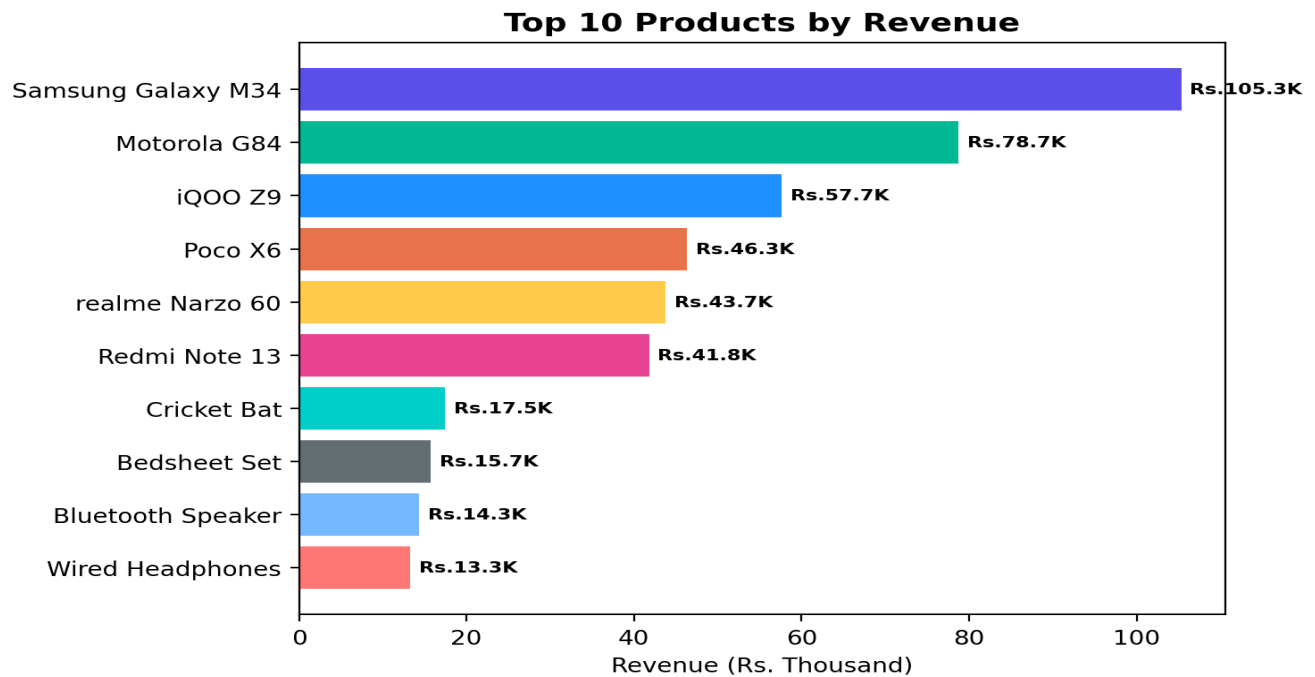
4.3 Order Status Distribution



- **Delivered** is the most common outcome at 64% (64 orders), the healthy baseline for a marketplace of this size.
- Cancelled (12) and Returned (14) orders together represent 26% of volume — worth tracking as a combined fulfilment-risk metric.

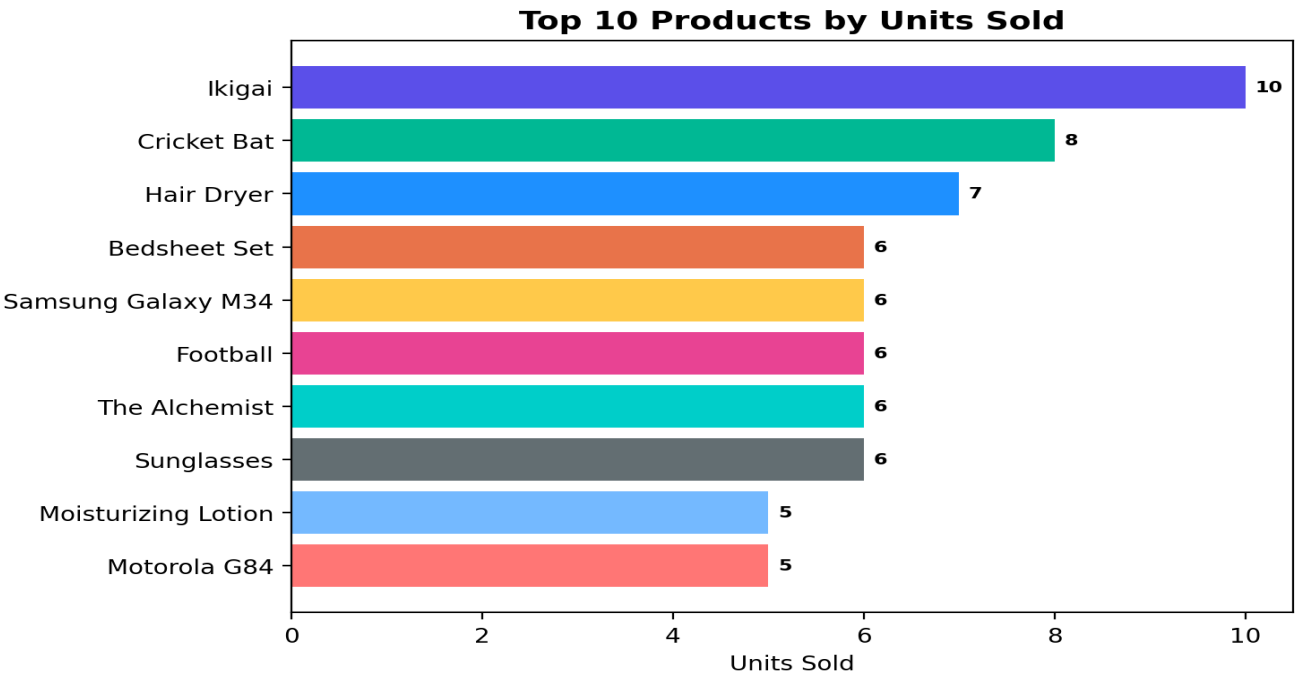
5. Product Performance

5.1 Top 10 Products by Revenue



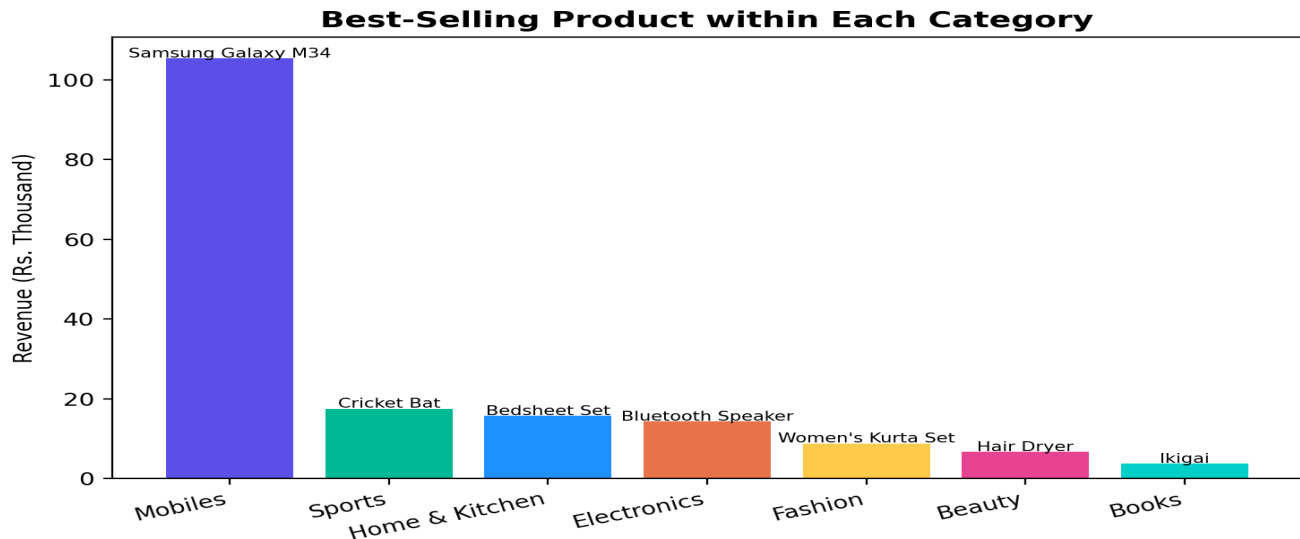
Samsung Galaxy M34 is the #1 revenue-generating product at Rs. 105,328. High-ticket Mobiles and Electronics items dominate the top of this chart even at low unit counts, since a single smartphone order can outweigh several Books or Beauty orders combined.

5.2 Top 10 Products by Units Sold



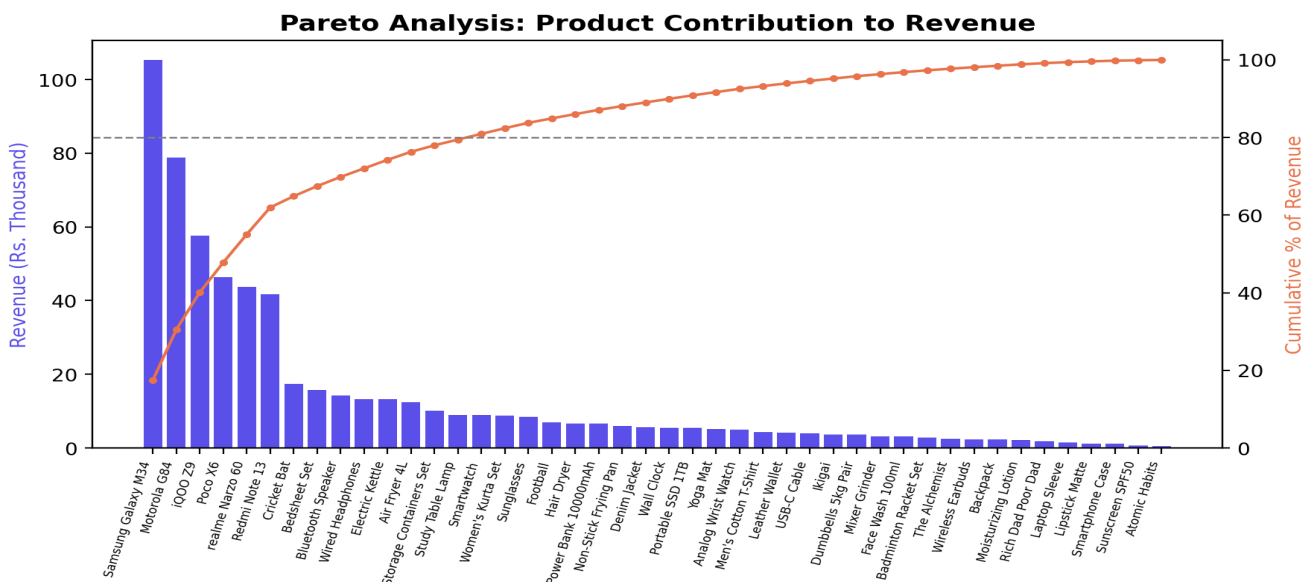
Ranking by units sold (rather than revenue) matters differently to warehousing than it does to finance — a lower-priced item outselling a higher-priced one in volume changes fulfilment priorities even if it doesn't top the revenue chart. Here, **Ikigai** leads with 10 units sold, reflecting repeat, lower-ticket purchases across Books and Sports categories.

5.3 Best-Selling Product within Each Category



Every category has a clear "hero SKU" that outsells its category peers — useful for prioritizing which single products to keep in stock first if a category-wide supply constraint ever hits.

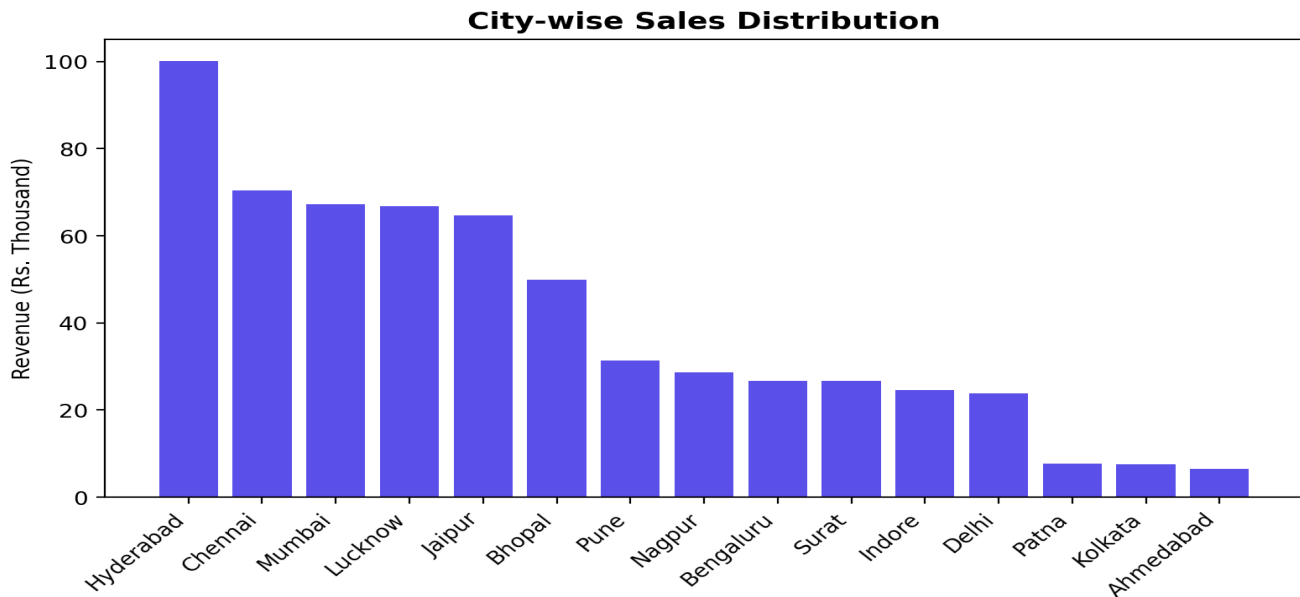
5.4 Pareto Analysis: Revenue Concentration



Just **15 of the platform's 44 products** account for roughly 80% of total revenue (the dashed line marks the 80% threshold on the cumulative curve). This is the classic "hero SKU" or Pareto pattern common in marketplaces — a pattern worth flagging directly to inventory planning, since a stock-out on any of these ~15 products would have an outsized revenue impact compared to a stock-out on a long-tail item.

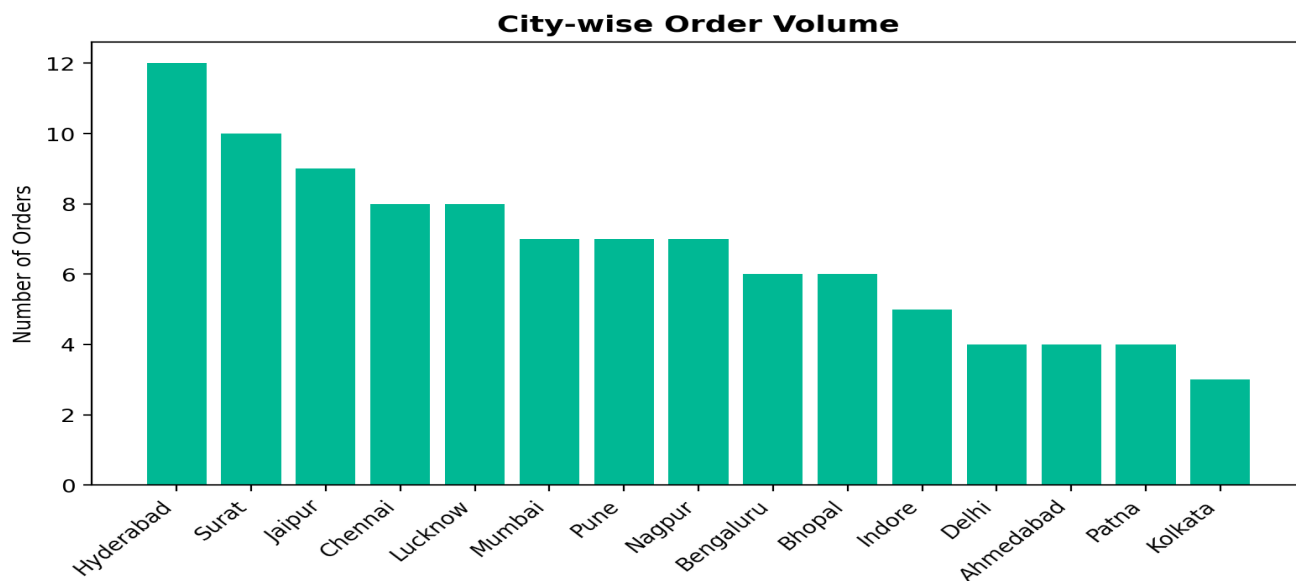
6. Geographic Demand

6.1 City-wise Sales Distribution



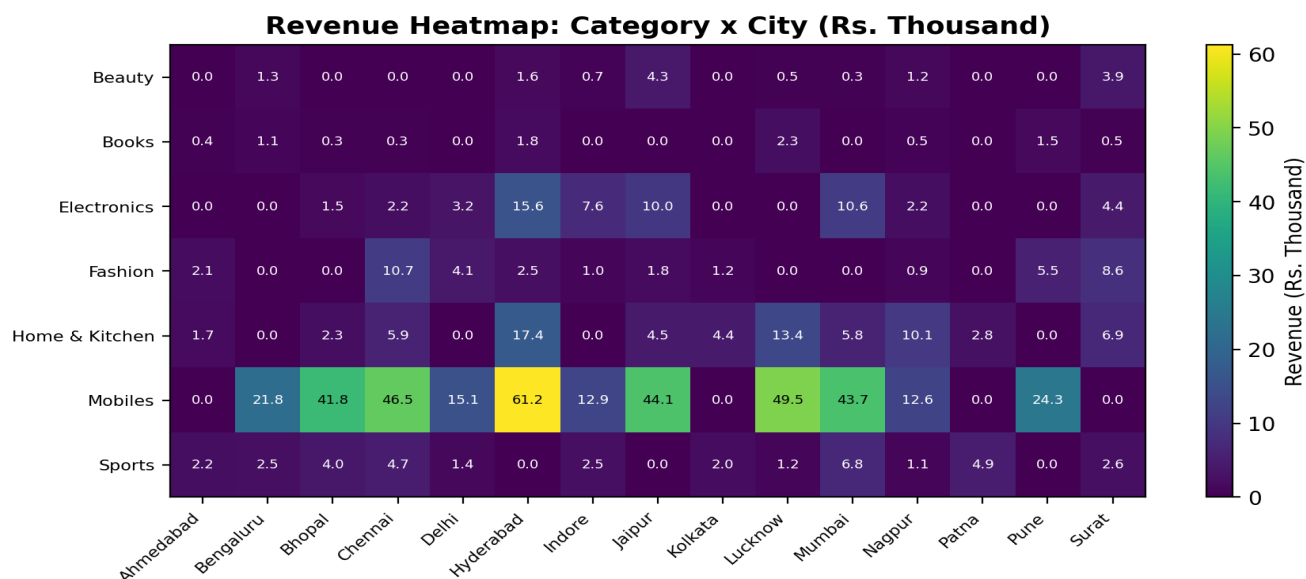
- **Hyderabad** leads all cities with Rs. 100,030 in sales.
- Revenue is spread across a wide set of Tier-1 and Tier-2 cities (Hyderabad, Chennai, Mumbai, Lucknow, Jaipur), indicating broad geographic penetration rather than concentration in a single metro.
- **Ahmedabad** recorded the lowest sales (Rs. 6,484), a candidate for localized marketing pushes.

6.2 City-wise Order Volume



Order-count ranking by city broadly mirrors the revenue ranking in 6.1, though a few cities with higher-value Mobiles/Electronics orders punch above their order-count weight in revenue terms.

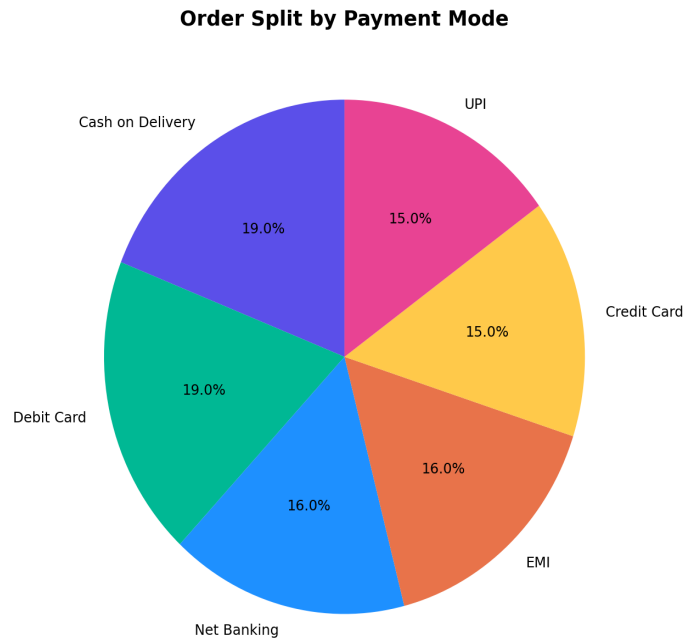
6.3 Revenue Heatmap: Category x City



The heatmap (values in Rs. thousand) surfaces category-city combinations that don't show up in either single-axis view alone — useful for spotting which cities are under-indexed on a given category and could be targeted with localized catalogue promotion.

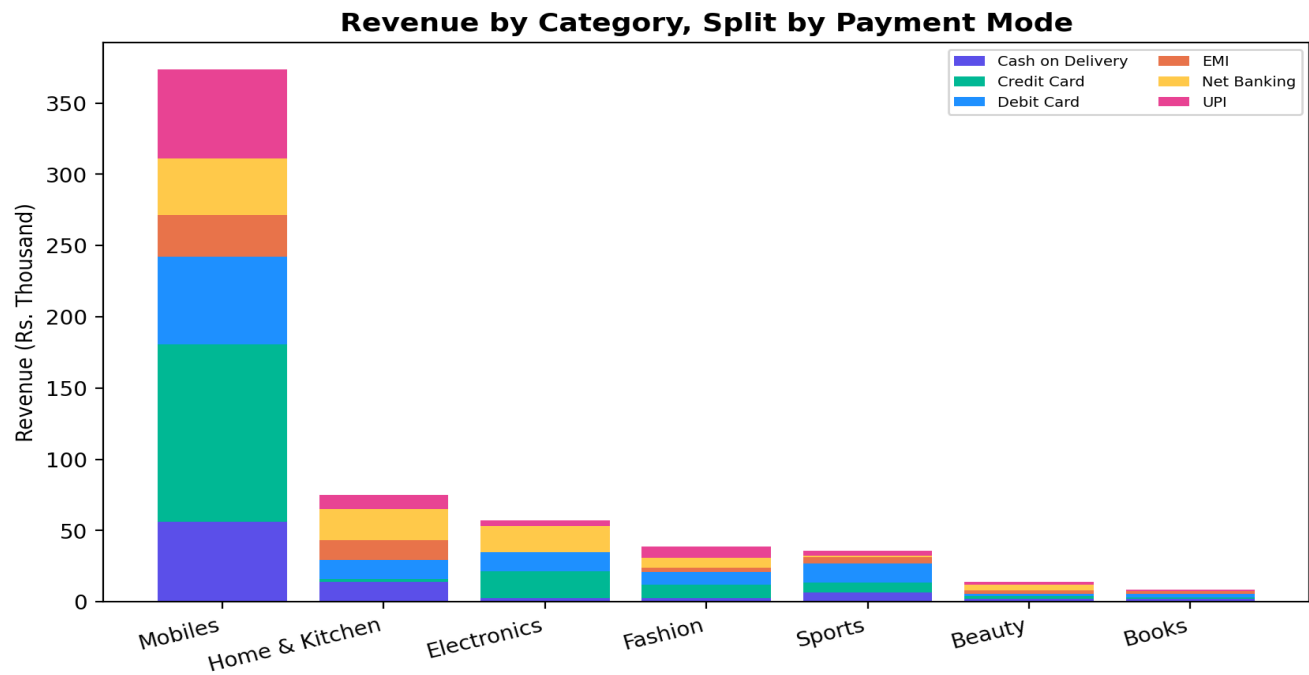
7. Payment Mode Analysis

7.1 Order Split by Payment Mode



- **Cash on Delivery** is the most-used payment mode, accounting for 19% of all orders.
- No single payment mode overwhelmingly dominates the mix — customers spread fairly evenly across UPI, cards, EMI, Net Banking, and Cash on Delivery, reflecting the varied price points in the catalogue (low-cost Books paid via UPI/COD vs. high-ticket Mobiles often financed via EMI or cards).

7.2 Revenue by Category, Split by Payment Mode



Breaking payment mode out by category shows where financing options matter most: high-ticket categories like Mobiles and Electronics lean more heavily on EMI and Credit Card, while low-ticket categories like Books and Beauty see a larger share of UPI and Cash on Delivery — a pattern worth reflecting in checkout-page payment-option ordering per category.

8. Order Fulfilment Analysis

8.1 Fulfilment Funnel

Status	Orders	% of Total
Delivered	64	64%
Shipped (in transit)	10	10%
Cancelled	12	12%
Returned	14	14%

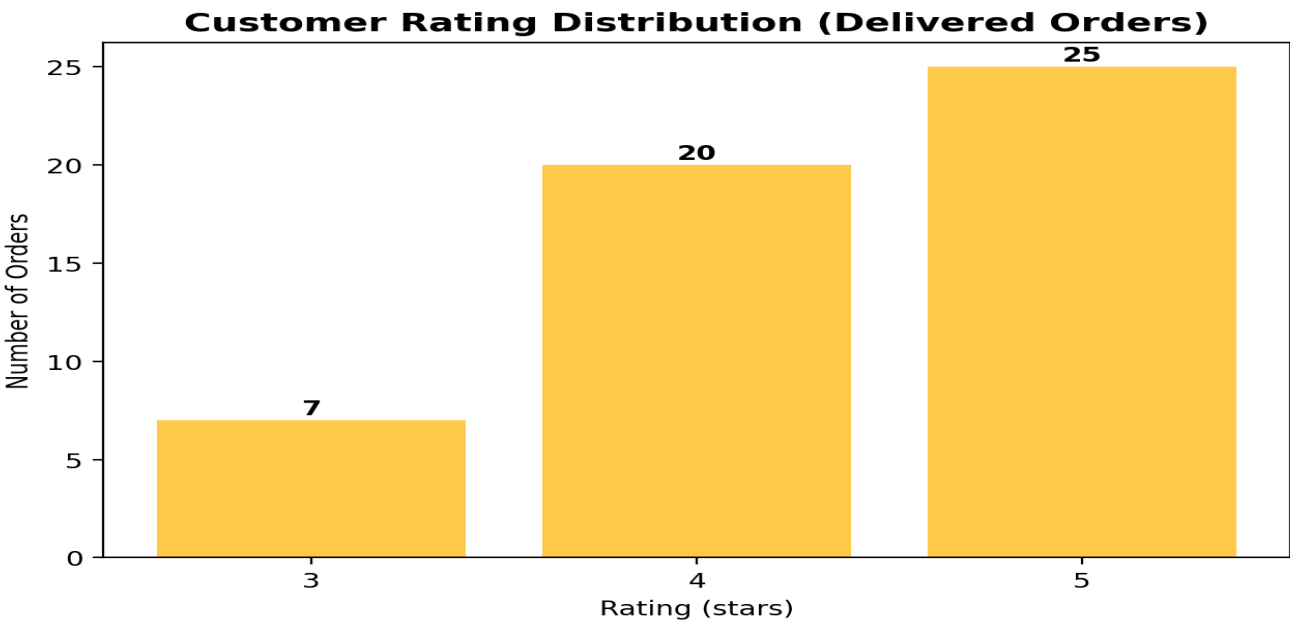
Of 100 total orders, 64% reached the customer successfully. Combined, Cancelled and Returned orders represent a meaningful share of lost fulfilment value — the next layer of analysis (customer-level return reasons, not available in this dataset) would help pinpoint whether losses concentrate in specific categories or cities.

8.2 What This Means Operationally

- Cancelled orders (before shipment) point to checkout-stage friction — payment failures, price shock, or stock mismatches — and are usually the cheapest fulfilment problem to fix.
- Returned orders (after delivery) are the costliest to resolve, involving reverse logistics — these deserve root-cause tracking by category, especially Fashion and Electronics where fit/spec mismatches are common industry-wide.
- Shipped orders still in transit at the time of this snapshot are excluded from both success and failure counts, since their final outcome is not yet known.

9. Customer Ratings Analysis

9.1 Rating Distribution (Delivered Orders Only)



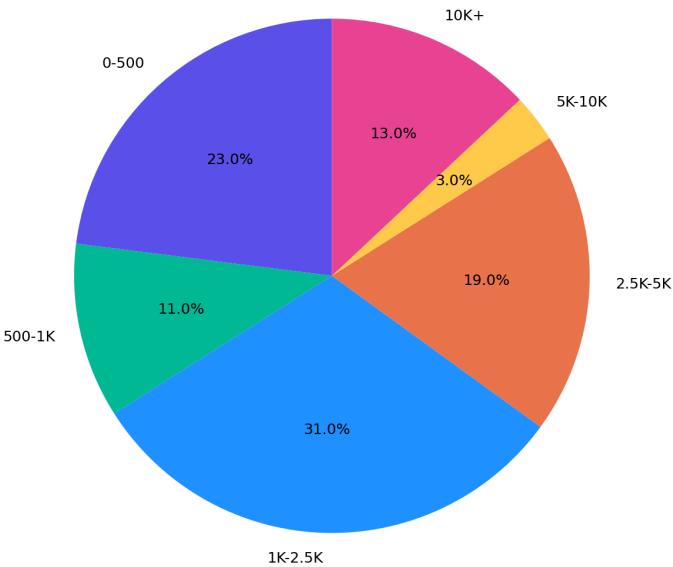
Average customer rating across all delivered orders is **4.35 out of 5**. Ratings are only captured for orders that reached the customer, so this metric reflects post-delivery satisfaction rather than overall order experience (which would also need to account for the Cancelled/Returned share in Section 8).

- A concentration of 4- and 5-star ratings suggests generally positive product-quality perception among customers who completed their purchase.
- Any 1- or 2-star ratings, even if rare in this sample, warrant category-level drill-down in a live dashboard to catch quality issues early.

10. Pricing Analysis

10.1 Order Distribution by Unit Price Band

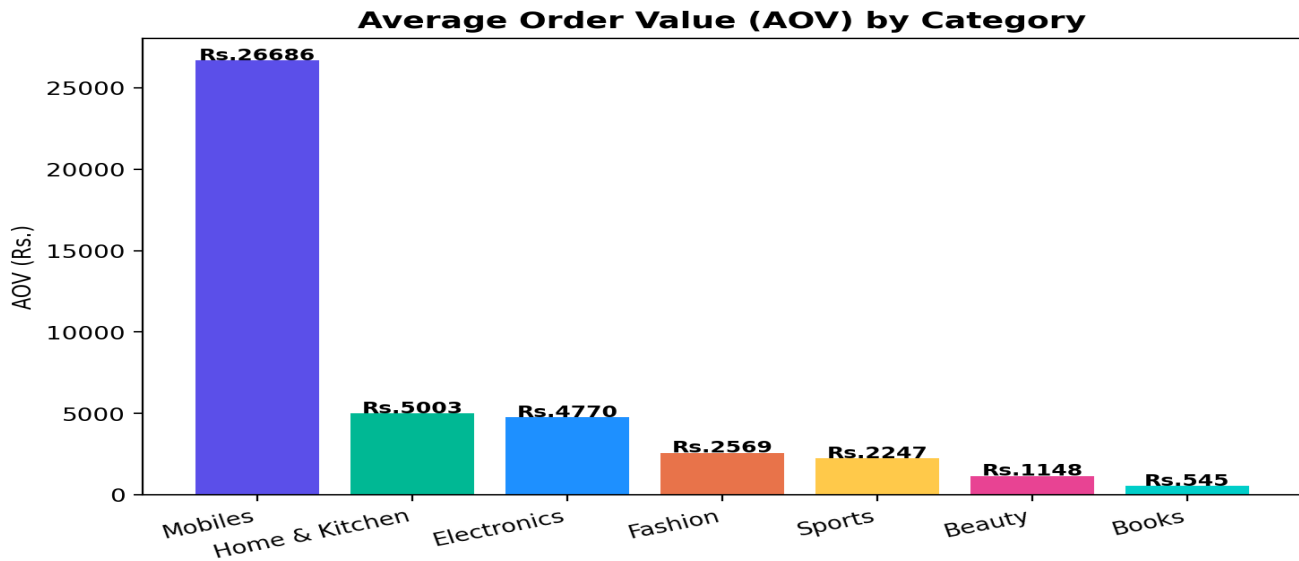
Order Distribution by Unit Price Band (Rs.)



Orders span a wide range of unit price bands from budget Books and Beauty items to premium Mobiles and Electronics — the catalogue mix supports both high-frequency, low-ticket purchases and occasional high-value transactions, which is typical of a general-merchandise marketplace.

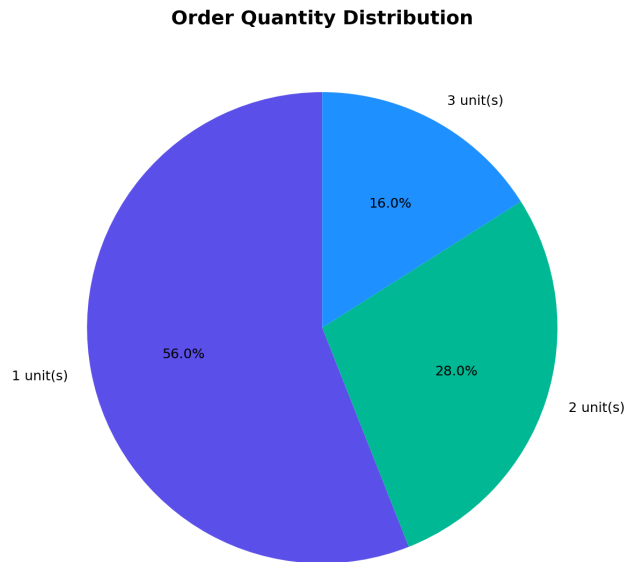
11. Order & Basket Behaviour

11.1 Average Order Value (AOV) by Category



- **Mobiles** has the highest AOV at Rs. 26,686 per order, driven by high-ticket smartphones.
- **Books** has the lowest AOV (Rs. 545), consistent with smaller, lower-cost individual purchases.

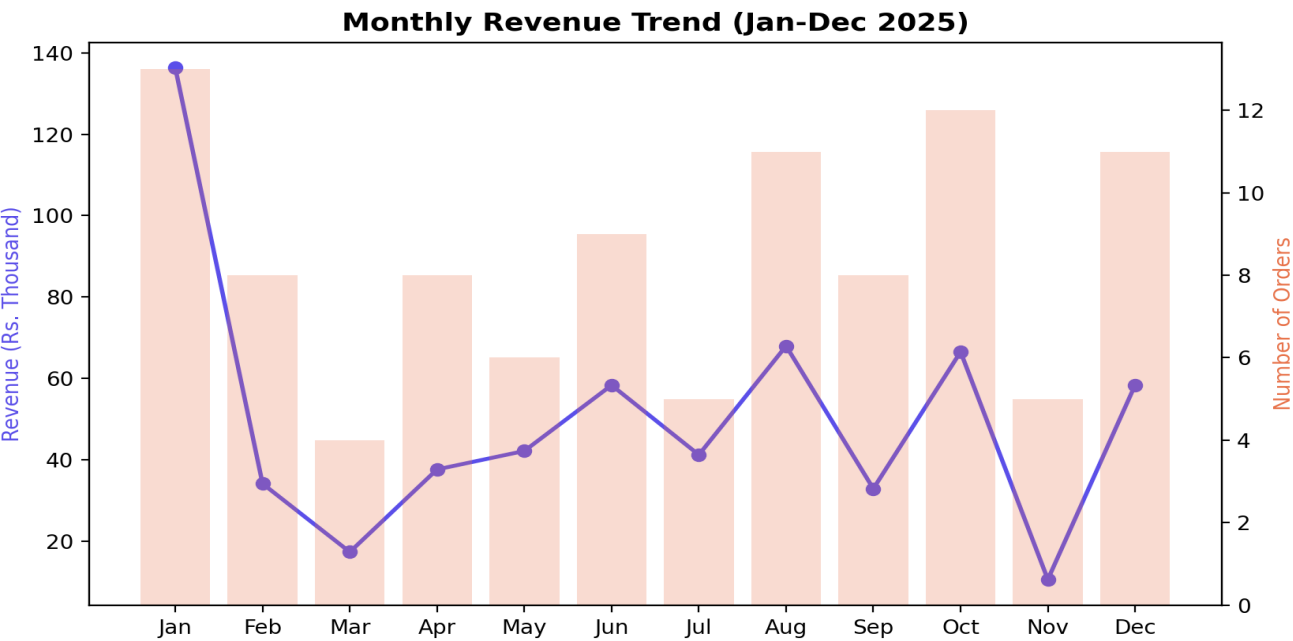
11.2 Order Quantity Distribution



1 unit(s) per order is the most common basket size, representing 56% of all orders. Multi-unit orders remain a minority — a possible upsell opportunity via bundle deals or quantity-based promotions.

12. Time Trend Analysis

12.1 Monthly Revenue Trend (2025)



Unlike a corrupted-date export, this dataset retained clean order dates — enabling a full month-over-month view. **Jan** was the peak month at Rs. 136,286 in revenue. Monthly revenue fluctuates across the year without a single dominant season, consistent with a general-merchandise catalogue rather than a strongly seasonal one (e.g., no single festive-quarter spike dominates the whole year in this sample).

13. Key Takeaways & Recommendations

- **Double down on Mobiles:** it drives the largest share of revenue — prioritize inventory depth, marketing spend, and new SKU launches here first.
- **Protect the ~15 hero SKUs:** 80% of revenue rides on a small subset of products (Section 5.4). These should carry the tightest stock-out monitoring and fastest replenishment priority.
- **Tighten the fulfilment funnel:** Cancelled and Returned orders together represent a meaningful 26% of volume (Section 8) — root-causing these by category and city should be a near-term priority.
- **Tailor payment options by category:** EMI and Credit Card usage skews toward high-ticket categories like Mobiles (Section 7.2) — surfacing these options earlier in checkout for such categories could reduce cart abandonment.
- **Target low-revenue cities:** Cities at the bottom of the sales chart (Section 6.1) are candidates for localized ads, regional promotions, or delivery-speed improvements.
- **Introduce bundle/multi-buy offers:** Most orders are single-unit (Section 11.2) — bundle pricing could raise AOV, especially in Books and Beauty where repeat, low-ticket purchases are common.
- **Monitor ratings by category:** With an average delivered-order rating of 4.35/5 (Section 9), tracking this metric per category would help catch emerging quality issues before they show up in return rates.

14. Next Steps: Descriptive → Predictive

This report answers what happened across 100 historical orders. The natural next layer, if useful, moves from descriptive to predictive and prescriptive analytics:

- Demand forecasting per SKU/city to set automatic reorder points, especially for the hero SKUs identified in the Pareto analysis.
- Return/cancellation prediction models that flag high-risk orders at checkout, using category, payment mode, and price band as features.
- Cohort / repeat-purchase analysis, if customer-level identifiers become available — this dataset is order-level only, with no customer ID field, so retention cannot currently be measured.
- Discount and promotion elasticity modelling, layering in campaign data if it becomes available, to right-size future markdowns by category.
- Live dashboard refresh — the same chart structure and section breakdown above would apply directly to a continuously refreshing data source.

14.1 How This Would Work at Real Scale

Layer	Purpose	Typical Tools
Ingestion	Capture each order event (purchase, payment, status change) in real time	Kafka / Kinesis from checkout & payments service
Stream processing	Roll up raw events into per-product, per-city, per-hour aggregates	Spark / Structured Streaming / Flink
Storage	Hold raw events (audit/reprocessing) and pre-aggregated tables (OLAP queries)	Data Lake (S3) or Data Warehouse (Redshift/Snowflake/BigQuery)
Batch aggregation	Nightly rollups: category revenue, fulfilment rates, city rankings, Pareto analysis	Agg. Parallelised SQL / dbt jobs
Serving / BI layer	Dashboards refreshed hourly or daily for category managers	Looker / Tableau / QuickSight

15. Appendix: Data Tables

15.1 Full Category Revenue Table

Category	Revenue (Rs.)	Orders	% of Total
Mobiles	Rs. 373,608	14	62.0%
Home & Kitchen	Rs. 75,047	15	12.4%
Electronics	Rs. 57,240	12	9.5%
Fashion	Rs. 38,540	15	6.4%
Sports	Rs. 35,946	16	6.0%
Beauty	Rs. 13,781	12	2.3%
Books	Rs. 8,722	16	1.4%

15.2 Full City Revenue Table

City	Revenue (Rs.)	Orders
Hyderabad	Rs. 100,030	12
Chennai	Rs. 70,329	8
Mumbai	Rs. 67,272	7
Lucknow	Rs. 66,801	8
Jaipur	Rs. 64,697	9
Bhopal	Rs. 49,962	6
Pune	Rs. 31,359	7
Nagpur	Rs. 28,688	7
Bengaluru	Rs. 26,733	6
Surat	Rs. 26,720	10
Indore	Rs. 24,652	5
Delhi	Rs. 23,901	4
Patna	Rs. 7,663	4
Kolkata	Rs. 7,593	3
Ahmedabad	Rs. 6,484	4

15.3 Full Product Revenue Table (all products)

Product	Revenue (Rs.)	Product	Revenue (Rs.)
Samsung Galaxy M34	Rs. 105,328	Wall Clock	Rs. 5,546
Motorola G84	Rs. 78,750	Portable SSD 1TB	Rs. 5,425
iQOO Z9	Rs. 57,670	Yoga Mat	Rs. 5,078
Poco X6	Rs. 46,337	Analog Wrist Watch	Rs. 4,967
realme Narzo 60	Rs. 43,730	Men's Cotton T-Shirt	Rs. 4,336
Redmi Note 13	Rs. 41,793	Leather Wallet	Rs. 4,149
Cricket Bat	Rs. 17,463	USB-C Cable	Rs. 3,908
Bedsheet Set	Rs. 15,679	Ikigai	Rs. 3,697
Bluetooth Speaker	Rs. 14,292	Dumbbells 5kg Pair	Rs. 3,669
Wired Headphones	Rs. 13,271	Mixer Grinder	Rs. 3,133
Electric Kettle	Rs. 13,243	Face Wash 100ml	Rs. 3,112
Air Fryer 4L	Rs. 12,500	Badminton Racket Set	Rs. 2,845
Storage Containers Set	Rs. 10,100	The Alchemist	Rs. 2,543
Study Table Lamp	Rs. 8,929	Wireless Earbuds	Rs. 2,311
Smartwatch	Rs. 8,862	Backpack	Rs. 2,239
Women's Kurta Set	Rs. 8,770	Moisturizing Lotion	Rs. 2,167
Sunglasses	Rs. 8,385	Rich Dad Poor Dad	Rs. 1,907
Football	Rs. 6,891	Laptop Sleeve	Rs. 1,512
Hair Dryer	Rs. 6,676	Lipstick Matte	Rs. 1,242
Power Bank 10000mAh	Rs. 6,560	Smartphone Case	Rs. 1,099
Non-Stick Frying Pan	Rs. 5,917	Sunscreen SPF50	Rs. 584
Denim Jacket	Rs. 5,693	Atomic Habits	Rs. 575

16. Closing Notes

16.1 Methodology Recap

Revenue = Unit Price x Quantity, summed per row. Category, product, city, and payment-mode splits use exact grouping on their respective source columns — no fuzzy matching or reconstruction was required since the source export preserved clean field boundaries throughout. Rating figures are scoped to Delivered orders only, since ratings are not collected for Cancelled, Returned, or in-transit Shipped orders. All monetary figures are in Indian Rupees (Rs.) and reflect order totals (unit price x quantity), not post-return net revenue.

16.2 Limitations

- No customer-level identifier exists in this dataset, so repeat-purchase and cohort/retention analysis could not be performed.
- No discount or MRP field exists (unlike some marketplace exports), so discount-leakage analysis was out of scope for this report.
- Return reasons and cancellation reasons are not captured, limiting root-cause analysis of the 26 non-delivered orders to descriptive counts only.

Ready when you are

Happy to build out any of the predictive extensions above, or re-run this entire report against a larger or live Flipkart data feed if one becomes available — the same chart structure and section breakdown above would apply directly.

Report generated from the Flipkart Sales Dataset (100 transactions, sample data for demo/practice purposes). All figures are descriptive summaries of the provided sales data.